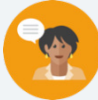


13.4 Partitioning Line Segments

Lesson Introduction

 Benchmark	MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.		 Learning Target	I can use coordinate geometry to partition a line segment.
 Benchmark Coherence	Building On MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line, and goes through a given point.		Working Toward N/A	
 Essential Questions	- How can you use coordinate geometry to partition a line segment? - What is a partitioned line segment?			
 Lesson Narrative	Students are exploring how to find a point that is used to partition a directed line segment by a given ratio. The Exploration part of the lesson guides students into finding ratios of partitions of lines. Students construct a conjecture that will be used in later parts of the lesson. Students will also relate the midpoint to partitioning line segments.			
 Component Summary	13.4.1 Warm-Up (~5 mins.)	Heroic person for students (SE p. 300)	13.4.3 Guided Instruction (15 mins.)	Practice problems on partitioning line segments (SE p. 302)
	13.4.2 Exploration (20 mins.)	Guided Discovery on partitioning directed line segments by a given ratio (SE p. 300)	13.4.4 Try It (5 mins.)	Two questions on partitioning line segments (SE p. 304)
	13.4.5 Wrap Up (~5 mins.)	Students summarize their learning (SE p. 305)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		N/A	
			Additional Preparation	
			If necessary, prepare visuals for students before the lesson.	

 Teacher Tips	Common Misconceptions • Students may struggle with visualizing part:part and part:whole ratios. • Students may struggle with writing ratios and identifying the denominators throughout the lesson. • Students may struggle with determining which endpoint belongs to which part of the ratio.	Things to Consider • Consider providing visuals for students to demonstrate the concept of partitioning. • Students should not be required to memorize a formula. Relating the formula to student conjectures and to similar triangles will help students in understanding why the formula works. • If time is an issue, then consider assigning 13.4.4 Try It for homework.
	Literacy and Language Routines	Math Thinking and Reasoning (MTR) Standard
	Compare and Connect Allow students to use the two methods of partitioning a directed line segment: the algebraic representation that is derived in the lesson, and the visual representation using slope. Have students make connections between the two methods, and have students demonstrate why both methods work.	MA.K12.MTR.2.1: Multiple Representations Students use graphs and equations to partition directed line segments, emphasizing the use of multiple representations. Having students use both an algebraic and a visual representation of partitioning a segment helps students make connections between the two representations.
 Differentiate Instruction	Consider using visuals to demonstrate how to partition a segment by using such graphing utilities as Desmos and GeoGebra. Allowing students to see visuals of partitioning may help students with determining the whole:part and part:part ratios in a directed line segment.	
Lesson Notes:		

13.4.1 Warm-Up: Heroic Person

Component Overview: Students answer two questions regarding a personal hero.



13.4.1 Warm-Up: Heroic Person

1. Who do you consider heroic and inspirational?
Answers will vary.
2. Why did you pick this person?
Answers will vary.



Consider having a few students who feel comfortable to share their responses with the class. Also, consider taking the time to read student choices, and write a short note to each student about their hero.

13.4.2 Exploration: Partitioning a Line Segment

Component Overview: Students will begin working with a line segment and will use ratios to partition a directed line segment into a given ratio.



13.4.2 Exploration: Partitioning a Line Segment

1. Directed line segment $\overline{SP}$ is shown.



- How many equal parts is $\overline{SP}$ partitioned into?
5
- Make a mark where $\overline{SP}$ is partitioned into a 2:3 ratio. Call this point R.
See above.
- Complete the statement.

Point R is

- ☐ $\frac{2}{3}$
☒ $\frac{2}{5}$
☐ $\frac{3}{5}$

of the distance from S to P.

- Complete the equation that can be used to find the distance of $\overline{SR}$.

$$SR = \frac{2}{5} SP$$



This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas and questions that you can ask them to share in the whole-group 13.4.3 Guided Instruction.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.



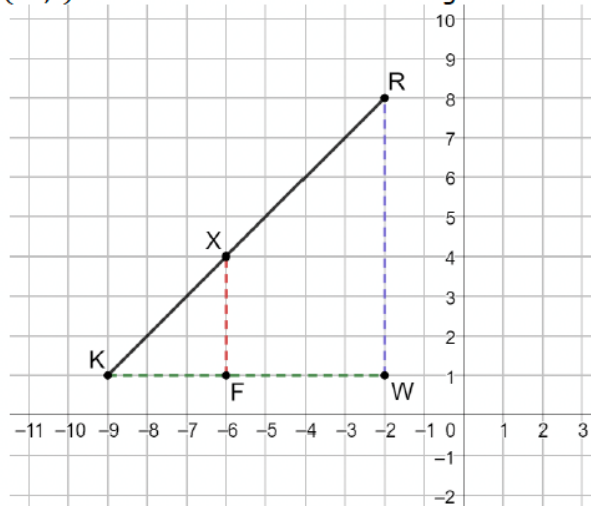
- What does it mean to partition something?
- How can the number line be used to determine a partition?



At first, students may forget to include all parts of the segment as the denominator. For instance, some students may struggle to conceptualize that point R is $\frac{2}{5}$ the distance between points S and P.

13.4.2 Exploration: Partitioning a Line Segment (continued)

2. Point X partitions directed line segment KR in a 3:4 ratio. Points $K(-9,1)$, $F(-6,1)$, $X(-6,4)$, $R(-2,8)$, and $W(-2,1)$ are shown on the coordinate grid.



- a. Use the distances to write the numeric ratio for each segment ratio. Then, choose the type of ratio the segment ratio represents.

Segment Ratio	Numeric Ratio	Type of Ratio
$KF:FW$	3:4	☐ part:part ☐ part:whole
$KF:KW$	3:7	☐ part:part ☐ part:whole
$XF:RW$	3:7	☐ part:part ☐ part:whole
$KX:KR$	3:7	☐ part:part ☐ part:whole

- b. What do you notice about the numeric ratios?
Answers will vary. The part:part ratio is different than the part:whole ratios. The part:whole ratios are all the same.



For a visual approach, have students label the lengths of all segments that are dotted in the graph.



Students may struggle with finding the part:part and the part:whole ratios. Having students label the lengths is important. In $KX:KR$, show students that the line goes through the coordinate plane evenly at each point, and that makes finding the ratio more accessible for students.



- How is part to part different from part to whole?
- How would the ratio change if the letters of the segments were reversed?

13.4.2 Exploration: Partitioning a Line Segment (continued)

- c. Two students were discussing the directed line segment KR .

Nevaeh	Isabella
Point R is a dilation of point X centered at point K using a scale factor of $\frac{3}{4}$.	Point R is a dilation of point X centered at point K using a scale factor of $\frac{3}{7}$.

Work with your partner to explain why neither student has the correct scale factor. Include in your explanation what the correct scale factor should be so that the result of dilating at point K is point R .

Nevaeh tried to apply the part:part ratio when she should have used the part:whole ratio. Isabella recognized to use the part:whole ratio, but the scale factor is the reciprocal of what she used. The correct scale factor is $\frac{7}{3}$.

- d. Discuss with your partner which ratios can be used to justify that $\triangle KXF \sim \triangle KRW$. Summarize your discussion.

Answers will vary. The ratios of the lengths of corresponding sides are all the same.

- e. With your partner, write a conjecture on how point X could be found if only given the endpoints of the directed line segment KR and the ratio.

Answers will vary.

$$x\text{-coordinate of } X: -9 + \frac{3}{7}(-2 - -9) = -6$$

$$y\text{-coordinate of } X: 1 + \frac{3}{7}(8 - 1) = 4$$

- f. Ask two other pairs of partners for the conjecture they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

Conjecture	Initials
<i>Answers will vary.</i>	



For a kinesthetic approach for collaboration for this activity, have students divide into two sides of the classroom, one side for Nevaeh, and the other for Isabella. Have each group discuss why neither student is correct on finding the scale factor. Then, have each group pick a spokesperson to share the group's findings.



As a segue to 13.4.3 Guided Instruction, have a few students share their conjectures with the class to be able to see any commonalities and differences.

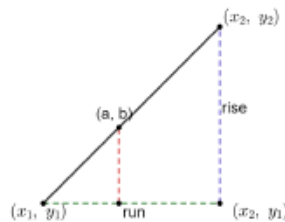
13.4.3 Guided Instruction: Partitioning a Line Segment

Component Overview: Teachers solidify student understanding from 13.4.2 Exploration by bringing together student conjectures and apply the conjectures to solve problems.



13.4.3 Guided Instruction: Partitioning a Line Segment

- Generalize the conjecture in the Exploration if the given ratio is part:whole.

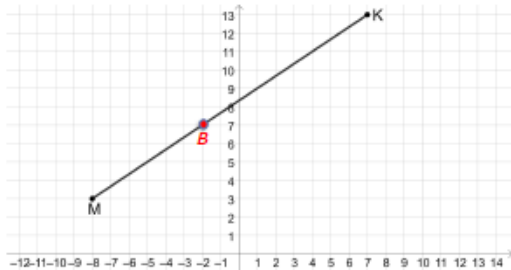


Generalization:

x-coordinate: $a = x_1 + (\text{part/whole})(\text{run})$

y-coordinate: $b = y_1 + (\text{part/whole})(\text{rise})$

- Directed line segment MK is shown with endpoints at point $M(-8,3)$ and point $K(7,13)$.



Directed line segment MK is partitioned by point B in a ratio of 2:3.

- Write an equation that can be used to find the distance of $\overline{MB}$.

$$MB = \frac{2}{5}MK$$



Consider beginning this section by discussing student conjectures, common errors, and student observations from 13.4.2 Exploration. Relating the formula to student conjectures and to similar triangles will help them understand why the formula works. Students should not be required to memorize a formula. Some students will prefer to use slope or some other form of the formula without writing the formula. Some students may prefer an equivalent way of writing the generalized form of the conjecture. If you have more than one class, the conjectures may vary.



- How would the equation change if the ratio is part:part?
- How do you know which endpoint of the line segment belongs to each part of the ratio?



Question 2 breaks down the generalization so that students experience the generalization. After teaching through all five parts, refer to the conjecture and point out how each part of the question relates to the generalization.

13.4.3 Guided Instruction: Partitioning a Line Segment (continued)

- b. Use your equation to find the horizontal distance of $\overline{MB}$.
 $MB = \frac{2}{5}(15)$; Horizontal distance = 6
- c. Use your equation to find the vertical distance of $\overline{MB}$.
 $MB = \frac{2}{5}(10)$; Vertical distance = 4
- d. Use the horizontal distance and the vertical distance to plot point B on $\overline{MK}$.
 See above.
- e. What are the coordinates of point B ?
 $(-2, 7)$
3. Directed line segment XY has endpoints at point $X(-6, 1)$ and point $Y(4, 4)$.



Directed line segment XY is partitioned $\frac{1}{2}$ of the distance from X to Y at point W .

- a. What is the horizontal distance of $\overline{XW}$?
 $\frac{10}{2} = 5$
- b. What is the vertical distance of $\overline{XW}$?
 $\frac{3}{2} = 1.5$
- c. What are the coordinates of point W ?
 $(-1, 2.5)$



Have students use different colored pens/pencils to label the horizontal and vertical distance for questions 2 and 3.



- Why is point W the midpoint?
- How is a line segment partitioned $\frac{1}{2}$ of the distance different from a line segment partitioned in a ratio of 1:2?

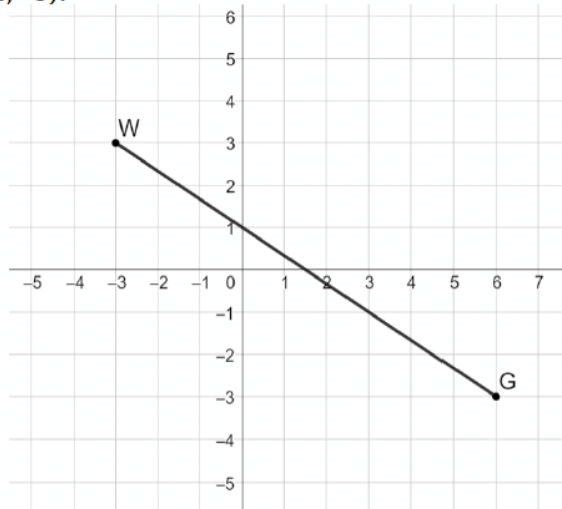
13.4.4 Try It: Partitioning a Line Segment

Component Overview: Students continue practicing how to find the point on a directed line segment by partitioning into a given ratio.



13.4.4 Try It: Partitioning a Line Segment

- Directed line segment WG has endpoints at $W(-3, 3)$ and $G(6, -3)$.



Directed line segment WG is partitioned $\frac{2}{3}$ of the distance from W to G at point D .

- What is the horizontal distance of $\overline{WD}$?

$$9 \left(\frac{2}{3} \right) = 6$$

- What is the vertical distance of $\overline{WD}$?

$$6 \left(\frac{2}{3} \right) = 4$$

- What are the coordinates of point D ?

$$(3, -1)$$



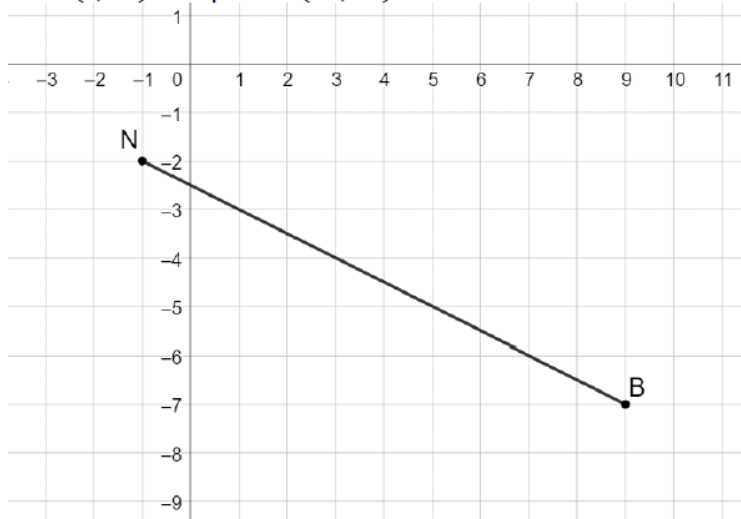
How can slope be used to find point D ?



If students finish this section early, have students create a directed line segment, and have students partition by several different ratios.

13.4.4 Try It: Partitioning a Line Segment (continued)

2. Directed line segment $\overrightarrow{BN}$ has endpoints at point $B(9, -7)$ and point $N(-1, -2)$.



Consider having some students use slope to find the coordinate of point H , and having some students use the generalization derived earlier in the lesson to find point H .

Directed line segment $\overrightarrow{BN}$ is partitioned by point H at a 4:1 ratio.

What are the coordinates of point H ?

Horizontal distance: $10 \left(\frac{4}{5} \right) = 8$

Vertical distance: $5 \left(\frac{4}{5} \right) = 4$

Point H : $(9 - 8, -7 + 4) \rightarrow (1, -3)$

13.4.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



13.4.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.